Who Bargains Next?

Recognition and cooperative allocations

Adrian Concord

Two anonymous bargaining protocols produce different patient allocations whenever pair worths differ. They share coalition opportunities, discounting, and proposer-departure risk; only the next proposer after a surviving rejection changes. Each protocol has a unique payoff vector across all subgame-perfect equilibria, with immediate efficient agreement. We derive the exact protocol effect and characterize when it vanishes.

Why it matters. The Shapley value divides a surplus according to what each party adds to every coalition, and it is the standard benchmark for fair division in cost sharing, coalition formation and attribution. Hart and Mas-Colell (1996) gave it a bargaining foundation: with random proposers and a risk that a rejected proposer drops out, patient players reach the Shapley value. The result below shows that this foundation depends on a rule the Shapley value never mentions: who proposes after a rejection. If the rejector proposes next, the patient outcome is $s + (1-q)(s-e)$ rather than s, where e is the equal split. Every player's deviation from an equal share is multiplied by $2-q$: players whose Shapley value exceeds one third gain, and players below one third lose. Giving the floor to whoever says no amplifies differences in bargaining power, and specifying the protocol after disagreement is part of specifying the outcome.

Model. There are three players with transferable utility, complete information, common discount factor $\delta \in (0,1)$, and characteristic function $v(\varnothing) = v(\{i\}) = 0$, $v(N) = 1$. Write $c_i = v(N \setminus \{i\})$, $C = \sum_i c_i$, $M = \max_i c_i$. Fix

$$\frac{3}{4} < q < 1, \qquad 0 < c_i, \qquad M < 2 - \frac{3}{2q}.$$

These inequalities imply monotonicity and supermodularity and define a nonempty open domain in the pair worths.

A newly active set draws a proposer uniformly. She offers any coalition of active players containing herself a nonnegative division of its worth. Respondents vote sequentially in a public uniform random order drawn after the offer. Acceptance pays coalition members immediately; they exit, and the residual set restarts next period with uniform recognition. A singleton receives zero. The first rejection ends voting. Next period, with probability q, the failed proposer exits for zero and the remaining set restarts uniformly. Otherwise everyone remains: protocol P_0 draws the next proposer uniformly, whereas P_1 gives that role to the rejector. Infinite nonagreement pays zero; there is no pass action. All agreements are voluntary. The only difference between protocols is recognition after a surviving rejection.

Let $\mathcal{E}_r(v,\delta)$ be expected payoffs before initial recognition in *all* subgame-perfect equilibria (SPE), including mixed and history-dependent strategies. Set

$$a = \delta(1-q), \quad b = \delta q, \quad d = \frac{b}{1-a}, \quad e = (1/3, 1/3, 1/3), \quad s_i = \mathrm{Sh}_i(v) = \frac{1}{3} + \frac{C - 3c_i}{6}.$$

Proposition 1. *Both protocols have SPE. Every SPE agrees immediately on the full active set whenever its worth is positive, and*

$$\mathcal{E}_0(v,\delta) = \{e + d(s-e)\}, \qquad \mathcal{E}_1(v,\delta) = \{e + d(1+a)(s-e)\}.$$

Their patient limits are s and $s + (1-q)(s-e)$. They coincide if and only if $c_1 = c_2 = c_3$. In any norm their Hausdorff distance is $da\|s-e\|$, converging to $(1-q)\|s-e\|$.

Proof. All continuation payoffs are nonnegative and sum to at most the active set's worth. In a pair of worth w, immediate threshold offers give proposer/responder payoffs $(w - aw/2, aw/2)$ under P_0 and $(w/(1+a), aw/(1+a))$ under P_1. These are SPE: rejection pays a proposer at most $aw < w(1-a)$. A proposer can obtain arbitrarily close to $w(1-a) > \max\{aw, 0\}$ by a strict offer, so singleton proposals are strictly inferior in every pair SPE. For unrestricted P_0 equilibria, let L_i, U_i bound i's pre-recognition payoff. Guarantees imply $2L_i \geq w - aU_j + aL_i$; feasibility gives $U_j \leq w - L_i$. Hence $L_i \geq w/2$, so both means equal $w/2$ and their conditional guarantees exhaust w. Under P_1, own-proposer extrema satisfy $L_i \geq w - aU_j$ and $U_i \leq w - aL_j$. Their widths contract by $a < 1$, fixing the displayed payoffs and immediate agreement. Thus every surviving pair gives each player $w/2$ before uniform recognition.

In a triple, offering each responder just above $a + bc_k/2$ guarantees acceptance. Therefore any current proposer k obtains at least

$$G = 1 - 2a - bM > 1/2 > \max\{a, M\}.$$

Every nonproposer obtains at most $1-G$. A proper-coalition offer pays its proposer at most M if accepted and a if rejected; it is strictly inferior. This excludes proper offers without assuming agreement.

For existence, let W_i^k be conditional payoffs, $H_i = W_i^i$, and $V_i = \frac{1}{3}\sum_k W_i^k$. For $r \in \{0,1\}$ solve

$$W_i^k = a[(1-r)V_i + rH_i] + bc_k/2 \quad (i \neq k), \qquad W_k^k = 1 - \sum_{i \neq k} W_i^k. \tag{1}$$

This map preserves the product of unit simplices, gives diagonals at least G, and contracts the maximum-entry norm by $2a < 1$. At its fixed point, a responder's own rejection is at least as valuable as the other responder's rejection, since $H_i > W_i^j$. Backward induction at every offer, with acceptance at equality on path, therefore implies that every accepted grand offer pays both responders at least their own-rejection thresholds. Such offers cannot improve the proposer's residual; rejection and proper offers pay strictly less. The pair strategies and discounted-tail argument complete an SPE.

To exclude other SPE under P_0, now let L_i, U_i be extrema before triple recognition, and $D_i = U_i - L_i$. Infima need not be attained. For proposer k and nonproposers i, j, any conditional payoff vector X^k satisfies

$$1 - a\sum_{h \neq k} U_h - bc_k \leq X_k^k \leq 1 - a\sum_{h \neq k} L_h - bc_k,$$
$$aL_i + bc_k/2 \leq X_i^k \leq aU_i + aD_j + bc_k/2.$$

The responder lower bound is an *expected* bound: an earlier rejection delivers it; otherwise her own rejection guarantees it. The proposer lower bound follows from strict offers, and the upper bounds from aggregate feasibility. Each row's coordinate widths are at most $a\sum_{h \neq k} D_h$. Averaging gives $D_i \leq (2a/3)\sum_h D_h$, so all widths vanish. The row bounds then collapse and sum to one. Any rejection has aggregate payoff at most $a + bM < 1$, forcing immediate efficient agreement.

Under P_1, take extrema conditional on being proposer. Own rejection gives a responder at least $aG + bc_k/2$; a later rejection by someone else gives at most $a(1-G) + bc_k/2$. Thus even with history dependence and mixed voting, every accepted branch pays each responder at least her own-rejection lower bound. Consequently

$$L_i \geq 1 - a\sum_{j \neq i} U_j - bc_i, \qquad U_i \leq 1 - a\sum_{j \neq i} L_j - bc_i.$$

Widths contract by $2a < 1$. The resulting fixed thresholds leave a unique proposer residual; rejection is strictly inferior, so every SPE agrees immediately and satisfies (1).

Finally put $t_i = a[(1-r)V_i + rH_i]$. Equation (1) gives $H_i = 1 - \sum_j t_j + t_i - bc_i$ and $3V_i = H_i + 2t_i + b(C - c_i)/2$. Subtraction yields $V_i - V_j = -d(1+ar)(c_i - c_j)/2$. Since $\sum_i V_i = 1$, the stated formula follows. Taking $\delta \uparrow 1$ gives $d \to 1$, $a \to 1-q$; the singleton Hausdorff distance and coincidence condition follow. □

Example and implication. Let $q = 7/8$ and $(c_1, c_2, c_3) = (1/4, 1/20, 1/20)$. The patient payoffs are

$$P_0 : \ (4/15,\ 11/30,\ 11/30), \qquad P_1 : \ (31/120,\ 89/240,\ 89/240).$$

Their maximum-coordinate distance is $1/120$, and separation persists on an open neighborhood of pair worths. Hence no protocol class containing these two procedures has a common singleton patient outcome on this domain. Equal initial proposal chances, anonymity, efficiency, and equilibrium payoff uniqueness do not remove the need to specify who controls bargaining after disagreement.

Relation to the literature. Gul (1989) establishes institutional sensitivity in coalitional bargaining. Hart and Mas-Colell (1996) use proposer departure; Calvo and Gutierrez-Lopez (2016) add time discounting. Kawamori (2008) studies rejector-contingent recognition, and Kawamori (2016) permits coalition offers with independent recognition. Britz, Herings, and Predtetchinski (2014) study rejector-dependent recognition with total breakdown. Here departure leaves a productive reduced game, its rule is held fixed, and the recognition comparison characterizes all SPE payoffs. We have not established whether this comparison appears elsewhere in the literature.

References. Britz, V., P. J.-J. Herings, and A. Predtetchinski (2014), "On the convergence to the Nash bargaining solution for action-dependent bargaining protocols," *Games and Economic Behavior* 86, 178–183. Calvo, E. and E. Gutierrez-Lopez (2016), "A strategic approach for the discounted Shapley values," *Theory and Decision* 80, 271–293. Gul, F. (1989), "Bargaining Foundations of Shapley Value," *Econometrica* 57, 81–95. Hart, S. and A. Mas-Colell (1996), "Bargaining and Value," *Econometrica* 64, 357–380. Kawamori, T. (2008), "A note on selection of proposers in coalitional bargaining," *International Journal of Game Theory* 37, 525–532. Kawamori, T. (2016), "Hart–Mas-Colell implementation of the discounted Shapley value," *Theory and Decision* 81, 357–369.